

Bluetooth Controlled Car Project

This project demonstrates a Bluetooth-controlled car using an HC-05 module, Arduino, and L298N motor driver.

Components:

Arduino Uno
HC-05 Bluetooth Module
L298N Motor Driver
2 DC Motors
Wheels & Chassis
Battery
Jumper Wires

```
// Bluetooth Controlled Car

char command;

int IN1 = 8;
int IN2 = 9;
int IN3 = 10;
int IN4 = 11;

void setup() {
  Serial.begin(9600);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
}

void loop() {
  if (Serial.available()) {
    command = Serial.read();

    if (command == 'F') {
      digitalWrite(IN1, HIGH);
      digitalWrite(IN2, LOW);
      digitalWrite(IN3, HIGH);
      digitalWrite(IN4, LOW);
    }
    else if (command == 'B') {
      digitalWrite(IN1, LOW);
      digitalWrite(IN2, HIGH);
      digitalWrite(IN3, LOW);
      digitalWrite(IN4, HIGH);
    }
    else if (command == 'L') {
      digitalWrite(IN1, LOW);
      digitalWrite(IN2, HIGH);
      digitalWrite(IN3, HIGH);
      digitalWrite(IN4, LOW);
    }
    else if (command == 'R') {
      digitalWrite(IN1, HIGH);
      digitalWrite(IN2, LOW);
      digitalWrite(IN3, LOW);
      digitalWrite(IN4, HIGH);
    }
    else if (command == 'S') {
      digitalWrite(IN1, LOW);
      digitalWrite(IN2, LOW);
      digitalWrite(IN3, LOW);
      digitalWrite(IN4, LOW);
    }
  }
}
```